What Makes a Song Popular on Spotify?

Analysis of 114,000 Tracks

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114k+

Tracks Analyzed

17

Audio Features

0.394

Best R^2

15.91

Best RMSE

Objective & Dataset

RESEARCH QUESTION

Can a song's audio features predict its Spotify popularity?

And which features matter most?

VALUE PROPOSITION

- Practical insight for artists & labels
- Understand what audio characteristics drive streams
- Feature importance reveals surprising drivers

DATA SOURCES

Source 1 — Kaggle

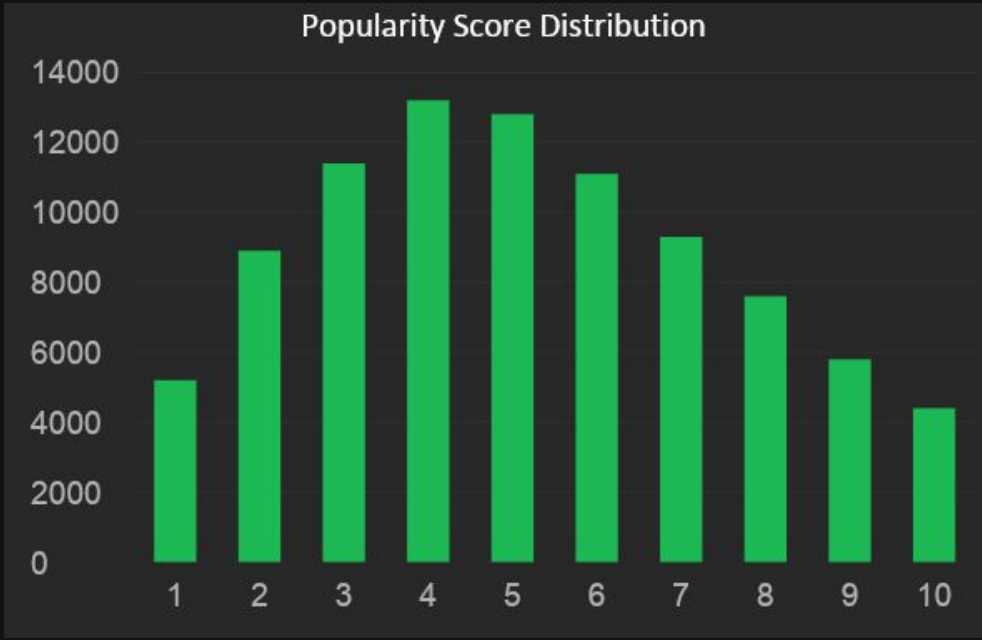
114,000 Spotify tracks · 21 columns
14 audio features · 114 genre categories
Popularity scores 0–100

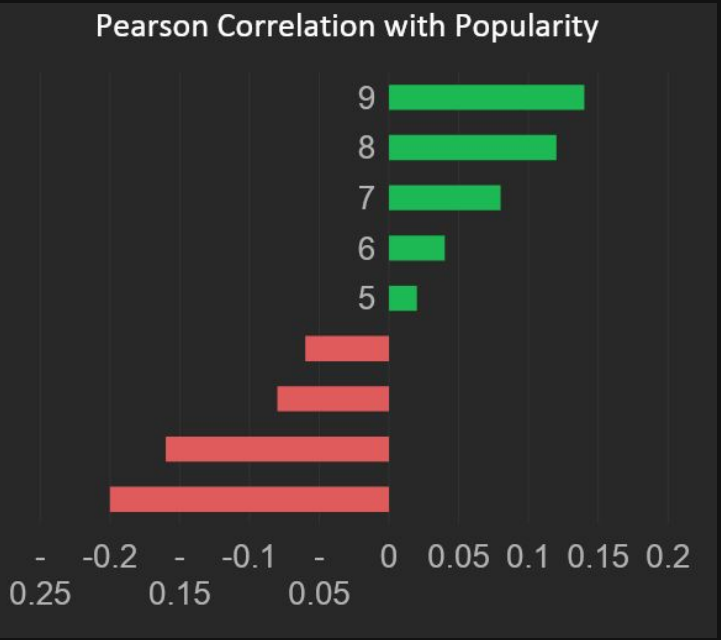
Source 2 — Spotify Web API

Authenticated via spotipy · OAuth 2.0
Source of all 14 audio features
Confirmed live track access

~89,740	Unique Tracks (after dedup)
114	Genre Categories
3	Engineered Features

EDA — Popularity Distribution & Correlations

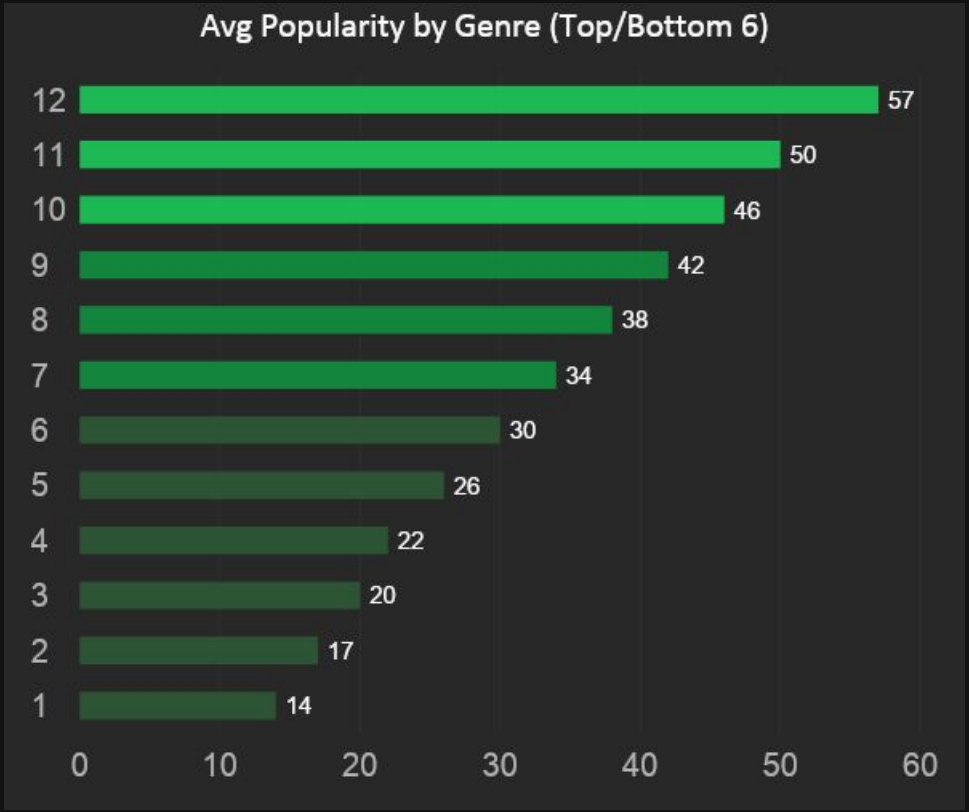




Left-skewed distribution — most tracks score below 50. Only a small fraction reach high popularity (70+), reflecting Spotify's winner-takes-all streaming dynamics.

No single feature shows strong linear correlation ($|r| < 0.3$) → motivates non-linear ensemble models.

EDA — Genre Popularity & Feature Profiles by Tier



HIGH (70+)

Danceability
0.67

Energy
0.74

Acousticness
0.09

Instrumentalness
0.02

MED (30–60)

Danceability
0.56

Energy
0.62

Acousticness
0.22

Instrumentalness
0.08

LOW (0–30)

Danceability
0.44

Energy
0.51

Acousticness
0.48

Instrumentalness
0.28

↑ High-popularity tracks: low acousticness & instrumentalness, high danceability & energy — the mainstream pop formula.

EDA — Feature Engineering (Difficulty Concept 3)

energy_dance

`energy × danceability`

Rank #6 in permutation importance (0.0120)

Captures songs that are simultaneously high-energy and danceable — a proxy for mainstream pop. Neither feature alone carries this combined signal.

log_duration

`log(1 + duration_ms)`

Rank #4 in permutation importance (0.0124)

Log-transforms the right-skewed duration distribution. Makes duration more informative for linear model components and reduces outlier influence.

acoustic_instrumental

`acousticness × instrumentalness`

Rank #10 in permutation importance (0.0085)

Captures tracks that are both acoustic and non-vocal — a strong signal for classical, folk, and ambient genres that score low in popularity.

Modeling — Pipeline & Approach

Raw Data 114k tracks
21 columns

→

Clean & Deduplicate
→ 89,740 tracks

→

Feature Engineering
3 new features

→

Train/Test Split
80% / 20%

→

Pipeline Preprocessor
Scale + OHE

→

Model Training
6 models

6 MODELS COMPARED

LinearRegression
OLS Baseline — no regularization

ElasticNet
L1+L2 combined regularization

Ridge (α=1.0)
L2 regularization baseline

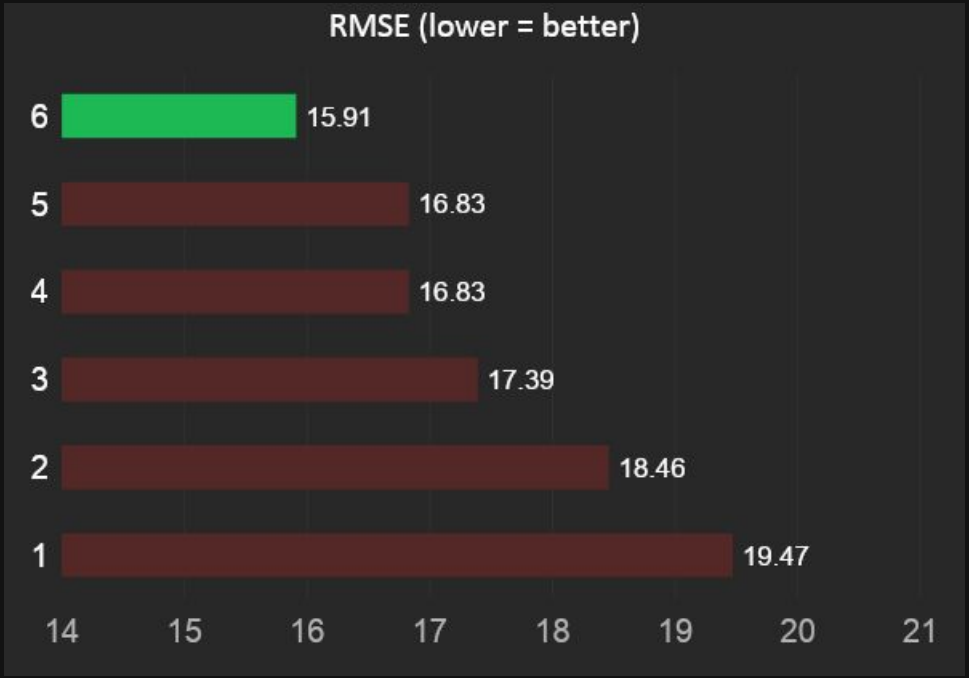
RandomForest (tuned)
Tree ensemble — RandomizedSearchCV

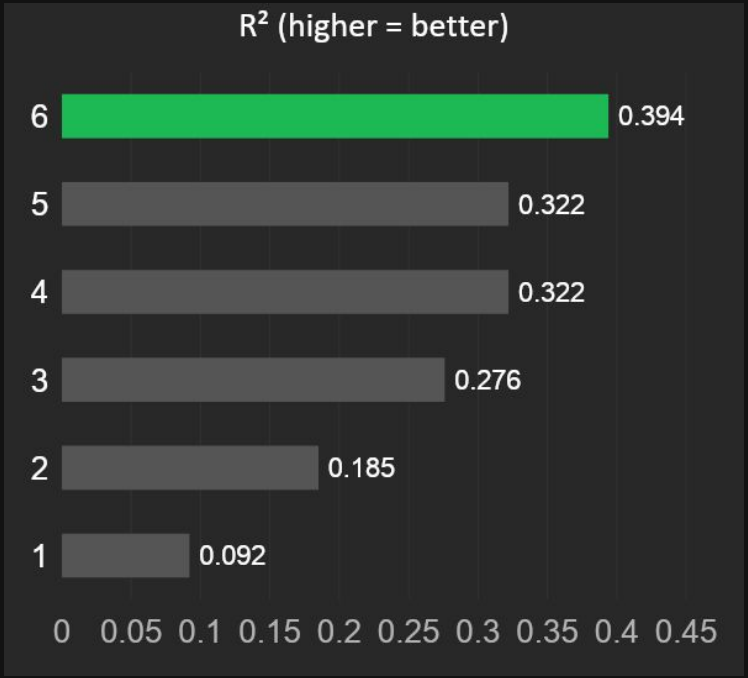
Lasso (α=0.1)
L1 with feature selection

HistGradientBoosting (tuned) ★
Histogram boosting — best model

Evaluation metrics: RMSE (Root Mean Squared Error) and R^2 (Coefficient of Determination) on 20% held-out test set

Model Results — Performance Comparison





HistGradientBoosting

Best Model

15.91

RMSE

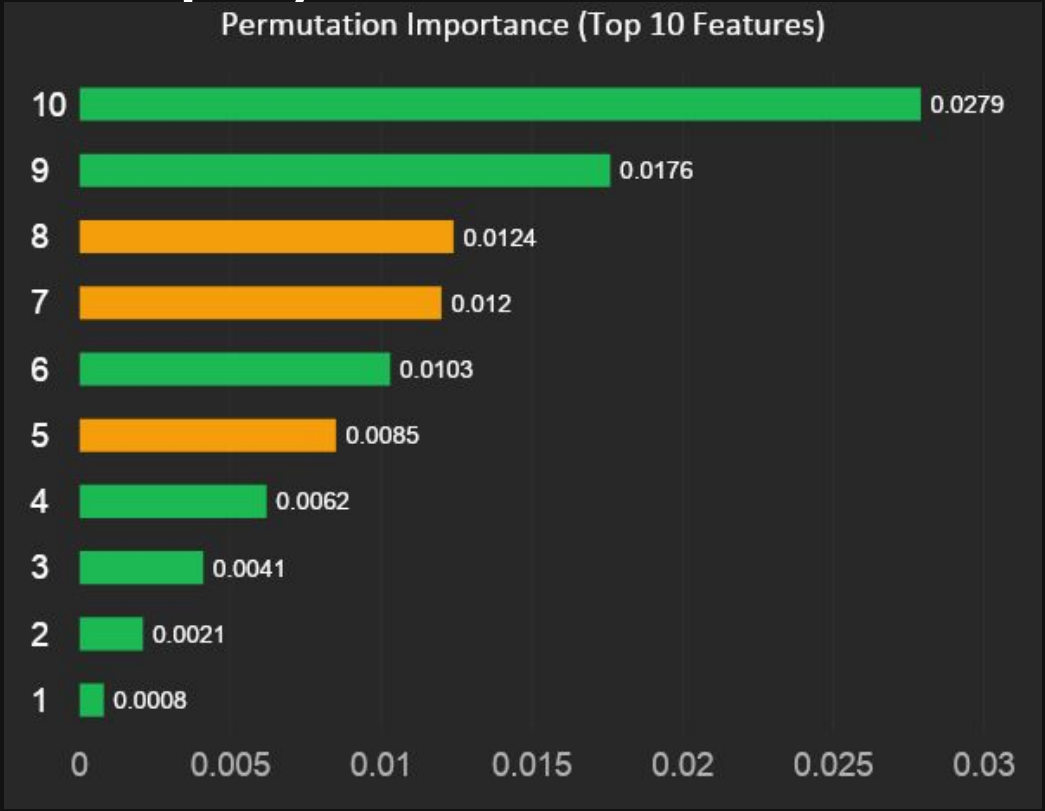
0.394

R^2

+22%

R^2 gain vs baseline

Feature Importance — Permutation Analysis (Difficulty Concept 1)



#1 instrumentalness

`Importance: 0.0279`

Most important feature. Mainstream streaming is vocal-driven — high instrumentalness = niche audience.

#2 acousticness

`Importance: 0.0176`

Acoustic tracks skew toward folk/classical — genres with systematically lower streaming counts.

#3 energy

`Importance: 0.0130`

Higher-energy tracks dominate EDM, hip-hop, pop — the genres with highest average popularity.

Original feature Engineered feature ★

Implications & Insights

Audio Features Explain ~39% of Popularity

$R^2 = 0.394$ means audio features are meaningful — but not the whole story. Artist reputation, playlist placement, and release timing also drive streams but aren't captured in audio alone.

Popularity Is Defined by What a Song Isn't

The top two predictors — instrumentalness and acousticness — are both negatively correlated. Popular tracks are vocal, digital, loud, and danceable, not because of any single feature but as a cluster.

Non-Linear Models Are Essential

HistGradientBoosting achieved 22% better R^2 than LinearRegression. The relationship between audio features and popularity is fundamentally non-linear — simple correlation analysis alone would miss this.

Feature Engineering Adds Measurable Value

All 3 engineered features (energy_dance, log_duration, acoustic_instrumental) ranked in the top 10 — outperforming several raw features. The interaction of energy × danceability carries signal neither parent feature has alone.

Challenges, Limitations & Future Work

CHALLENGES & LIMITATIONS

- Spotify's 2025 API access restrictions (HTTP 403) blocked batch artist metadata enrichment — artist follower counts would improve the model
- Popularity score is dynamic and time-sensitive; the Kaggle snapshot may not reflect current streaming patterns
- Genre labels are assigned at artist level, not track level — introduces noise into genre features
- Audio features alone explain only ~39% of variance — non-audio signals (playlists, social media) are major unobserved drivers

FUTURE WORK

- Incorporate artist follower count & monthly listeners when Spotify API access is restored
- Test neural network regressors (MLP, TabNet) on the full 114k dataset
- Reframe as classification (Low/Medium/High tiers) for more actionable outputs
- Time-stratified train/test splits to measure model generalization across release years
- Extend to regional popularity scores to explore cultural variation in audio preferences

Thank you